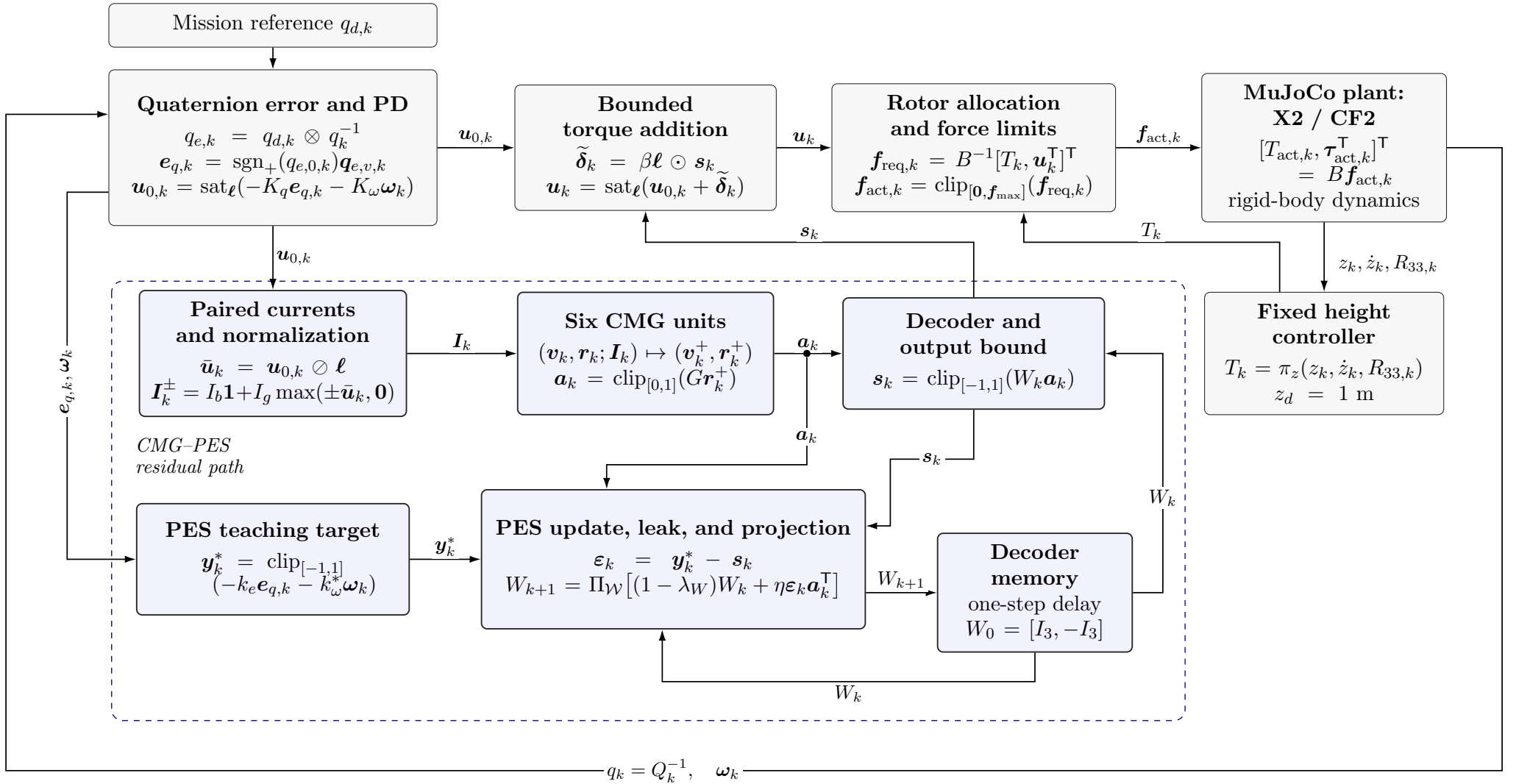

Mission reference $q_{d,k}$
Quaternion error and PD
$q_{e,k} = q_{d,k} \otimes q_k^{-1}$
$e_{q,k} = \mathrm{sgn}_+(q_{e,0,k}) q_{e,v,k}$
$u_{0,k} = \mathrm{sat}_\ell(-K_q e_{q,k} - K_\omega \omega_k)$
$u_{0,k}$
Bounded torque addition
$\tilde{\delta}_k = \beta \ell \odot s_k$
$u_k = \mathrm{sat}_\ell(u_{0,k} + \tilde{\delta}_k)$
u_k
Rotor allocation and force limits
$f_{\mathrm{req},k} = B^{-1}[T_k, u_k^\top]^\top$
$f_{\mathrm{act},k} = \mathrm{clip}_{[0, f_{\max}]}(f_{\mathrm{req},k})$
$f_{\mathrm{act},k}$
MuJoCo plant: X2 / CF2
$[T_{\mathrm{act},k}, \tau_{\mathrm{act},k}^\top]^\top = B f_{\mathrm{act},k}$
rigid-body dynamics
s_k
T_k
$z_k, \dot{z}_k, R_{33,k}$
Paired currents and normalization
$\bar{u}_k = u_{0,k} \oslash \ell$
$I_k^\pm = I_b \mathbf{1} + I_g \max(\pm \bar{u}_k, \mathbf{0})$
I_k
Six CMG units
$(v_k, r_k; I_k) \mapsto (v_k^+, r_k^+)$
$a_k = \mathrm{clip}_{[0,1]}(G r_k^+)$
a_k
Decoder and output bound
$s_k = \mathrm{clip}_{[-1,1]}(W_k a_k)$
Fixed height controller
$T_k = \pi_z(z_k, \dot{z}_k, R_{33,k})$
$z_d = 1$ m
$e_{q,k}, \omega_k$
CMG–PES residual path
PES teaching target
$y_k^* = \mathrm{clip}_{[-1,1]}(-k_e e_{q,k} - k_\omega^* \omega_k)$
y_k^*
PES update, leak, and projection
$\varepsilon_k = y_k^* - s_k$
$W_{k+1} = \Pi_{\mathcal{W}}[(1 - \lambda_W) W_k + \eta \varepsilon_k a_k^\top]$
W_{k+1}
Decoder memory
one-step delay
$W_0 = [I_3, -I_3]$
W_k
W_k
$q_k = Q_k^{-1}, \quad \omega_k$